

**National University of Computer and Emerging Sciences, Lahore Campus**



|              |                                                     |              |           |
|--------------|-----------------------------------------------------|--------------|-----------|
| Course Name: | Microprocessor and Emerging Sciences, Lahore Campus | Course Code: | EE303     |
| Program:     | BS(EE)                                              | Semester:    | Fall 2018 |
| Duration:    | 60 Minutes                                          | Total Marks: | 30        |
| Paper Date:  | 01-10-2018                                          | Weightage    | 15%       |
| Section:     | ALL                                                 | Page(s):     | 1         |
| Exam Type:   | Mid Term - I                                        |              |           |

Student Name: Hashim Raza

Roll No. 15-4521 Section: EE-2

**Instruction/Notes:** Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once.

**Question#1:**

Show the contents of all registers used and affected RAM locations after the execution of the following code. [10 Marks] CLO-01

```

ORG 0H
MOV R1, #40H
MOV R0, #40H
MOV R2, #3
MOV R3, #4
MOV DPTR, #LIST
AGAIN:
MOV A, 3H
MOVC A, @A+DPTR
MOV R4, A
MOV @R0, 4H
INC R0
INC R3
DJNZ R2, AGAIN
MOV A, @R1
ADD A, #2
PUSH ACC
INC R1
MOV A, @R1
ADD A, #2
PUSH ACC
HERE: SJMP HERE
LIST: DB 1, 28H, 7, 6, 10, 0CH, -2, 16, 10H
END
    
```

**Question#2:**

[10 Marks] CLO-02

Write an assembly language program to find RAM locations of all zero valued bytes in scratch pad of general purpose RAM (35H - 40H) of 89C52 microcontroller. The address of these RAM locations must be stored on Stack. The first element (to be stored) on Stack must be at 20H location.

**Question#3:**

[10 Marks] CLO-02

8051 is an 8-bit microcontroller that performs arithmetic on 8-bit numbers. However, we can manipulate 16 bits numbers as well. Write an assembly language program that stores two 16 bits numbers in the RAM such that lower byte of 1<sup>st</sup> number is stored in R0 and higher byte in R1. Similarly lower byte of 2<sup>nd</sup> number is stored in R2 and higher byte in R3. Now subtract these two 16 bit numbers and store lower byte of the result in accumulator A and higher byte in accumulator B.